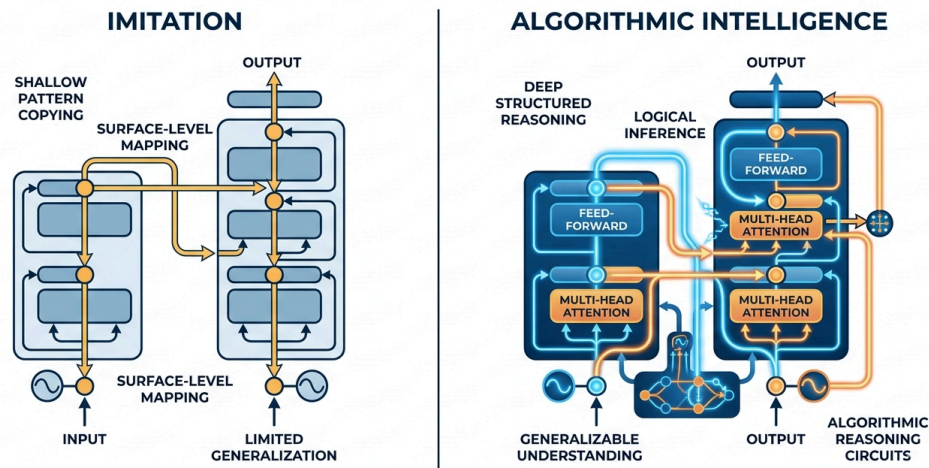


Introduction

Large Language Models exhibit remarkable In-Context Learning (ICL) — solving novel tasks from a handful of examples without any weight updates. But a deeper question lurks beneath this capability:

Does the model understand the rule, or is it just a very good mimic?
This work introduces Prompt-Level Grokking — the hypothesis that transformers undergo a sudden, discontinuous transition inside the context window: from surface imitation to true latent algorithm abstraction. This mirrors the delayed generalization ("grokking") observed during training, but now localized entirely within inference time.

Central Question: Does a transformer experience a "realization moment" inside the prompt itself — a threshold at which pattern copying collapses into genuine algorithmic reasoning?



Motivation and Problem Statement

Training-time grokking (Power et al., 2022) revealed that neural networks can suddenly generalize long after apparent convergence — a phase transition from memorization to understanding. The ask is: does an analogous phenomenon occur during inference?

- The ICL Grokking Hypothesis:
- Stage 1 — **Imitation**: Model copies example surface patterns
 - Stage 2 — **Partial Abstraction**: Latent rules begin forming in attention heads
 - Stage 3 — **Realization**: A critical prompt exposure triggers rule crystallization
 - Stage 4 — **Transfer**: Model generalizes to out-of-distribution rule instances

Why this matters for AGI: If transformers can cross from mimicry into rule abstraction within context, it suggests a primitive form of in-context "understanding" — a capability central to general intelligence. Understanding when and why this transition fires is a step toward understanding intelligence itself.

Objective

- Detect whether a phase transition exists from pattern-following to rule-understanding inside ICL
- Identify the critical prompt exposure that triggers abstraction
- Map the mechanistic circuits (induction heads, MLP neurons) active during the transition
- Test whether the transition resembles classical grokking dynamics (loss plateau → sudden jump)
- Probe whether post-transition behavior constitutes genuine algorithmic generalization — a hallmark of AGI-like reasoning

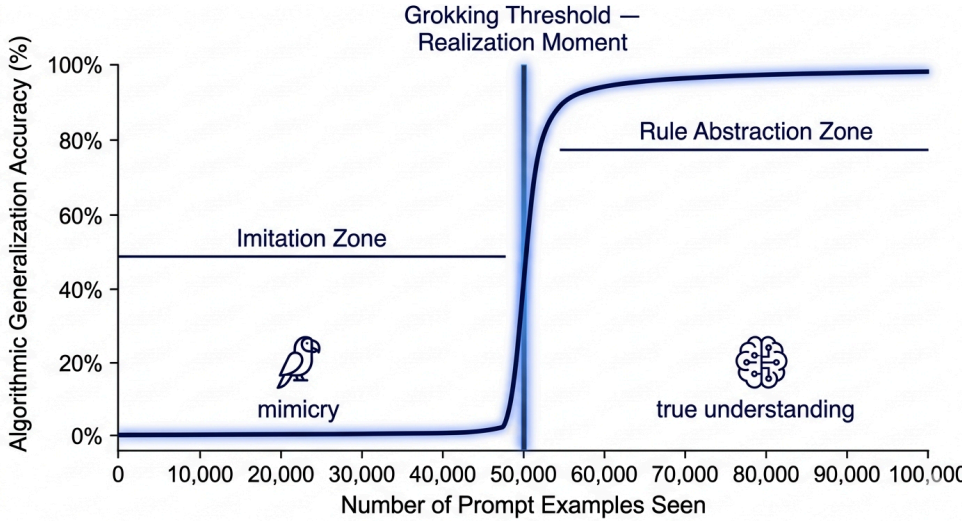
Conclusion

This work introduces **Prompt-Level Grokking** — a novel hypothesis proposing that transformers undergo a sharp, discontinuous transition within the context window itself, moving from surface-level pattern imitation to genuine latent algorithm abstraction. Situated at the intersection of grokking dynamics, in-context learning, and mechanistic interpretability, this framework reframes ICL not as mere retrieval or interpolation, but as a dynamic process capable of crossing a critical threshold into rule-based reasoning. If validated, this would suggest that intelligence-like behavior can emerge from structured prompt exposure alone — independent of training scale — and that transformers harbour latent circuits capable of self-activating into an algorithmic mode. Identifying when and why this transition fires offers a concrete, mechanistically grounded step toward understanding the emergence of general intelligence.

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[3] Akyürek, E., Schuurmans, D., Andreas, J., Ma, T., & Zhou, D. (2022). What Learning Algorithm Is In-Context Learning? Investigations with Linear Models. In Proceedings of ICLR 2023.
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Methodology

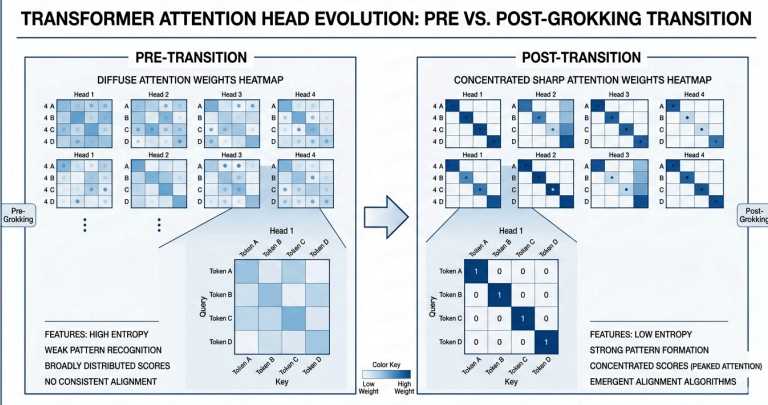


Task Families (where ground-truth rule learning is verifiable) Modular arithmetic · Sequence continuation · Symbolic rewriting · Graph traversal · Dynamic programming toy tasks

Prompt Curriculum (graduated complexity) Exact demonstrations → Structurally similar variants → Noisy distractors → Adversarial rephrasing → Compositional rule chains

- Evaluation Signals:**
- Accuracy jump profile (smooth vs. discontinuous)
 - Token-level confidence trajectory across prompt families
 - Attention concentration on induction vs. composition heads
 - Out-of-distribution transfer rate post-transition

Mechanistic Analysis
(mechanistic interpretability):
Using probing classifiers and activation patching to isolate:
induction head activation,
MLP neurons encoding latent rules,
scratchpad-like representations



Results

Phase	Behavior	Signal
Early prompts	Imitation	Low OOD accuracy, diffuse attention
Mid prompts	Partial rule formation	Induction head activation rises
Critical transition	Sudden accuracy jump	Attention concentrates, MLP fires
Post-transition	Algorithmic generalization	Robust to distractors, OOD transfer

Key prediction: The transition is not gradual — it mirrors the grokking curve, suggesting a universal mechanism of delayed abstraction independent of whether learning happens in weights or context.